Area 5: Mechanical Components and Assemblies

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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5.A Pressurized Vessels and Piping

1. Key concepts

- Thin wall (t < D/20): hoop σ_t = pD/(2t), longitudinal σ_L = pD/(4t). Sphere: σ = pD/(4t).
- Thick wall (Lamé): σ_t = [p_ir_i² − p_or_o² + r_i²r_o²(p_i − p_o)/r²]/(r_o² − r_i²). Internal pressure only: the maximum is at r_i, σ_t = p(r_o² + r_i²)/(r_o² − r_i²).
- ASME VIII Div. 1 shell: t = PR/(SE − 0.6P), with R the inside radius and E the joint efficiency.

2. Essential formulas

Name	Formula	Variables / units
Hoop	$\sigma_t = pD/(2t)$	
Longitudinal	$\sigma_L = pD/(4t)$	
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$	
ASME shell	$t = PR/(SE - 0.6P)$	

3. Handbook search keywords

thin-walled pressure vessel | Lamé thick-walled cylinder | ASME Section VIII thickness

4. Worked examples

Example 1: ASME shell thickness
R = 30 in, P = 250 psi, S = 20,000 psi, E = 0.85. Find t.

1. t = 250(30)/(20,000·0.85 − 0.6·250) = 0.445 in (add the corrosion allowance).

Answer: 0.445 in

5. Common traps

- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.A Pressurized Vessels and Piping: one-page cheat summary

► $\sigma_t = pD/2t$; $\sigma_L = pD/4t$

► Lamé max at the bore

► $t = PR/(SE - 0.6P)$

Formulas

Hoop	$\sigma_t = pD/(2t)$
Longitudinal	$\sigma_L = pD/(4t)$
Lamé (internal)	$\sigma_{t,max} = p(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$
ASME shell	$t = PR/(SE - 0.6P)$

Search keywords

thin-walled pressure vessel, Lamé thick-walled cylinder, ASME Section VIII thickness

Top traps

- Using the radius in pD/2t (D is the diameter).
- Using the thin-wall formula when t/r > 0.1.

5.B Hydraulic and Pneumatic Components

1. Key concepts

- Cylinder force F = pA. On retraction use the annulus A = π(D² − d²)/4. Speed v = Q/A.
- Hydraulic power: hp = Q(gpm)·p(psi)/1,714. Input = output/η.
- Pneumatics: the air is compressible, so use absolute pressure in gas-law calculations.

2. Essential formulas

Name	Formula	Variables / units
Force	$F = pA$	
Speed	$v = Q/A\ (231\ \text{in}^3/\text{gal})$	
Power	$hp = Q{\cdot}p/1{,}714$	

3. Handbook search keywords

hydraulic cylinder force · hydraulic power 1714 · pump efficiency

4. Worked examples

Example 1: Cylinder speed and force
A Ø3 in bore cylinder at 1,500 psi is supplied with 12 gpm. Find the extension force and speed.

1. A = 7.069 in²; F = 10,603 lb.
2. v = 12(231)/60/7.069 = 6.54 in/s.

Answer: F ≈ 10,603 lb; v ≈ 6.54 in/s

5. Common traps

- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.B Hydraulic and Pneumatic Components: one-page cheat summary

- ► $F = pA$
- ► $v = Q/A$
- ► $hp = Qp/1{,}714$

Formulas

Force	$F = pA$
Speed	$v = Q/A\ (231\ \text{in}^3/\text{gal})$
Power	$hp = Q{\cdot}p/1{,}714$

Search keywords

hydraulic cylinder force, hydraulic power 1714, pump efficiency

Top traps

- Using the full bore area on retraction.
- Gauge pressure in gas laws.

5.C Beams, Trusses, and Frames (Statics)

1. Key concepts

- Free-body diagram first: isolate the body, show every external force and moment (weight, supports, friction). Unknown reactions: pin = 2 forces, roller = 1, fixed = 2 forces + 1 moment.
- Equilibrium: ΣF_x = 0, ΣF_y = 0, ΣM = 0. Take moments about the point with the most unknowns.
- Trusses: method of joints (ΣF = 0 at each pin; start where there are ≤ 2 unknowns) or method of sections (cut ≤ 3 members, then take moments). Zero-force members: at an unloaded joint where two members are collinear, the third member is zero.
- Friction: F ≤ μ_sN. Impending motion means F = μ_sN. Belt/capstan: $T_1/T_2 = e^{(\mu\theta)}$.
- Centroids and moments of inertia of composites: $\bar{x} = \Sigma A\, x/\Sigma A$, then $I = \Sigma(I_c + Ad^2)$.

2. Essential formulas

Name	Formula	Variables / units
Equilibrium	$\Sigma F = 0\ ;\ \Sigma M = 0$	
Friction	$F = \mu N$ (impending)	
Capstan	$T_1 = T_2 e^{(\mu\theta)}$	θ rad
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$	
Parallel axis	$I = \Sigma(I_c + Ad^2)$	
Two-force member	force along the line joining its pins	

3. Handbook search keywords

equilibrium · method of joints · method of sections · zero-force member · friction · centroid · parallel axis

4. Worked examples

Example 1: Beam reaction
A 12 ft simple beam carries 9,000 lb at 4 ft from support A. Find R_A and R_B.

1. ΣM_B = 0: R_A(12) = 9,000(8).
2. R_A = 6,000 lb; R_B = 9,000 − 6,000 = 3,000 lb.

Answer: R_A = 6,000 lb, R_B = 3,000 lb

Example 2: Ladder friction
A uniform ladder leans at 60° from the floor against a smooth wall. Find the minimum μ at the floor.

1. Smooth wall: horizontal reaction N_w only. Floor: N = W, F = N_w.
2. ΣM about the foot: N_w·L sin60 = W(L/2)cos60 → N_w = W/(2 tan60).
3. μ_min = N_w/N = 1/(2 tan60°) = 0.289.

Answer: 0.289

Example 3: Truss member
Two members meet at an apex at 40° to the horizontal and carry 10,000 lb vertically. Find the force in each.

1. 2F sin40° = 10,000 → F = 7,779 lb (compression).

Answer: 7,779 lb

5. Common traps

- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.
- Using cos where sin belongs (draw the angle).
- Taking moments about a point where the unknowns do not drop out.

5.C Beams, Trusses, and Frames (Statics): one-page cheat summary

- ▶ FBD → $\Sigma F = 0, \Sigma M = 0$
- ▶ Pin 2, roller 1, fixed 3 unknowns
- ▶ Zero-force: 2 collinear + 1 at an unloaded joint
- ▶ $F = \mu N$ impending
- ▶ $T_1/T_2 = e^{\wedge}(\mu\theta)$
- ▶ $\bar{y} = \Sigma Ay/\Sigma A$; $I = \Sigma(I_c + Ad^2)$

Formulas

Equilibrium	$\Sigma F = 0$; $\Sigma M = 0$
Friction	$F = \mu N$ (impending)
Capstan	$T_1 = T_2 e^{\wedge}(\mu\theta)$
Centroid	$\bar{y} = \Sigma A_i y_i / \Sigma A_i$
Parallel axis	$I = \Sigma(I_c + Ad^2)$
Two-force member	force along the line joining its pins

Search keywords

equilibrium, method of joints, method of sections, zero-force member, friction, centroid, parallel axis

Top traps

- Leaving the friction force or the weight off the FBD.
- Adding a 'ma' force to a static FBD.
- Using cos where sin belongs (draw the angle).

5.D Springs

1. Key concepts

- Helical compression spring: $k = Gd^4/(8D^3N_a)$. Spring index $C = D/d$ (4–12 is good).
- Stress: $\tau = K_W \cdot 8FD/(\pi d^3)$, with Wahl $K_W = (4C - 1)/(4C - 4) + 0.615/C$ (or Bergsträsser $K_B = (4C + 2)/(4C - 3)$).
- Series springs: $1/k = \Sigma 1/k_i$. Parallel: $k = \Sigma k_i$. Energy $U = \tfrac{1}{2}kx^2$.
- Spring fatigue: Zimmerli data with torsional Goodman. Surge frequency: avoid resonance with the operating frequency.

2. Essential formulas

Name	Formula	Variables / units
Rate	$k = Gd^4/(8D^3N)$	
Stress	$\tau = K \cdot 8FD/(\pi d^3)$	
Wahl	$K = (4C - 1)/(4C - 4) + 0.615/C$	
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$	

3. Handbook search keywords

helical spring rate · Wahl factor · spring index · springs in series

4. Worked examples

Example 1: Spring rate
Music wire $G = 11.5\times10^6$ psi, $d = 0.148$ in, $D = 1.5$ in, $N_a = 8$. Find k.

1. $k = 11.5\times10^6(0.148)^4/(8\cdot1.5^3\cdot8) = 25.5$ lb/in.

Answer: 25.5 lb/in

5. Common traps

- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.D Springs: one-page cheat summary

- $k = Gd^4/8D^3N$
- $\tau = K8FD/\pi d^3$
- Series $1/k = \Sigma 1/k$

Formulas

Rate	$k = Gd^4/(8D^3N)$
Stress	$\tau = K8FD/(\pi d^3)$
Wahl	$K = (4C - 1)/(4C - 4) + 0.615/C$
Series/parallel	$1/k = \Sigma 1/k_i$; $k = \Sigma k_i$

Search keywords

helical spring rate, Wahl factor, spring index, springs in series

Top traps

- Using the outside diameter for D (use the mean coil diameter).
- Leaving out the curvature factor.

5.E Vibrating Systems

1. Key concepts

- SDOF: $\omega_n = \sqrt{(k/m)}$, $f_n = \omega_n/2\pi$. Static deflection shortcut: $\omega_n = \sqrt{(g/\delta_{st})}$, $f_n \approx 3.13/\sqrt{\delta}$(in) Hz.
- Damping: $c_c = 2\sqrt{(km)} = 2m\omega_n$, $\zeta = c/c_c$. Damped frequency $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$. Log decrement $\delta = (1/n)\ln(x_0/x_n)$, $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.
- Forced response: $X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, with $r = \omega/\omega_n$. Resonance is near $r = 1$.
- Transmissibility $TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$. Isolation ($TR < 1$) needs $r > \sqrt{2}$. Above $\sqrt{2}$, damping INCREASES TR, and softer springs or more mass help. Undamped: $TR = 1/(r^2 - 1)$.
- Rotating unbalance: $X = (m_e e/M)r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$.
- Equivalent systems: springs in series or parallel. Cantilever $k = 3EI/L^3$; simply supported center $k = 48EI/L^3$. Torsional $k_t = GJ/L$, $\omega_n = \sqrt{(k_t/J_{mass})}$. Compound pendulum $\omega_n = \sqrt{(mgd/I_O)}$.

2. Essential formulas

Name	Formula	Variables / units
Natural frequency	$\omega_n = \sqrt{(k/m)}\ ;\ f = \omega_n/2\pi$	
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$	
Damping ratio	$\zeta = c/(2\sqrt{(km)})$	
Damped	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$	
Log decrement	$\delta = (1/n)\ln(x_0/x_n)\ ;\ \zeta \approx \delta/2\pi$	
Amplitude	$X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$	
Transmissibility	$TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$	
Beam stiffness	$\text{cantilever } 3EI/L^3\ ;\ \text{SS center } 48EI/L^3$	

3. Handbook search keywords

natural frequency · damping ratio · logarithmic decrement · transmissibility · vibration isolation · rotating unbalance · forced vibration

4. Worked examples

Example 1: Damping and frequency
m = 10 kg, k = 100 kN/m, c = 150 N·s/m. Find f_n, ζ and f_d.

1. $\omega_n = \sqrt{(100{,}000/10)} = 100.0$ rad/s → $f_n = 15.92$ Hz.
2. $c_c = 2\sqrt{(km)} = 2{,}000$ N·s/m; $\zeta = 150/2{,}000 = 0.075$.
3. $f_d = f_n\sqrt{(1 - \zeta^2)} = 15.87$ Hz (practically the same).

Answer: $f_n = 15.92$ Hz, $\zeta = 0.075$

Example 2: Isolation
A machine on mounts runs at $r = 3$ with $\zeta = 0.1$. Find TR. Would more damping help?

1. $TR = \sqrt{[(1 + 0.6^2)/((1 - 9)^2 + 0.6^2)]} = 0.1454$.
2. No. Above $r = \sqrt{2}$, extra damping raises TR.

Answer: $TR \approx 0.145$

Example 3: Isolator static deflection
Undamped mounts must give TR = 0.20 at 1,800 rpm. Find the required static deflection.

1. $r^2 = 1/TR + 1 = 6 \rightarrow r = 2.449$.
2. $\omega = 1{,}800 \cdot 2\pi/60 = 188.5$ rad/s; $\omega_n = \omega/r = 77.0$ rad/s.
3. $\delta = g/\omega_n^2 = 386.1/5922 = 0.0652$ in.

Answer: 0.065 in

5. Common traps

- Mixing Hz and rad/s ($\omega = 2\pi f$).
- Thinking damping always helps isolation (it hurts above $\sqrt{2}$).
- Leaving k in kN/m or N/mm while m is in kg.
- Using the series formula for parallel springs.

5.E Vibrating Systems: one-page cheat summary

- ► $\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta)}$
- ► $\zeta = c/2\sqrt{(km)}$
- ► $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
- ► $\zeta \approx \ln(x_0/x_n)/(2\pi n)$
- ► TR < 1 only for $r > \sqrt{2}$
- ► Undamped $TR = 1/(r^2 - 1)$
- ► k: cantilever $3EI/L^3$, SS $48EI/L^3$

Formulas

Natural frequency	$\omega_n = \sqrt{(k/m)}$; $f = \omega_n/2\pi$
Static deflection	$\omega_n = \sqrt{(g/\delta_{st})}$
Damping ratio	$\zeta = c/(2\sqrt{(km)})$
Damped	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
Log decrement	$\delta = (1/n)\ln(x_0/x_n)$; $\zeta \approx \delta/2\pi$
Amplitude	$X = (F_0/k)/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$
Transmissibility	$TR = \sqrt{[(1 + (2\zeta r)^2)/((1 - r^2)^2 + (2\zeta r)^2)]}$
Beam stiffness	cantilever $3EI/L^3$; SS center $48EI/L^3$

Search keywords

natural frequency, damping ratio, logarithmic decrement, transmissibility, vibration isolation, rotating unbalance, forced vibration

Top traps

- Mixing Hz and rad/s ($\omega = 2\pi f$).
- Thinking damping always helps isolation (it hurts above $\sqrt{2}$).
- Leaving k in kN/m or N/mm while m is in kg.

5.F Basic Machines and Mechanisms (Dynamics)

1. Key concepts

- Kinematics: constant acceleration gives $v = v_0 + at$, $s = v_0t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2as$. Rotation is the same with θ, ω, α, and $v = r\omega$, $a_t = r\alpha$, $a_n = r\omega^2$.
- Kinetics: $\Sigma F = ma$ and $\Sigma M_G = I_G\alpha$. Draw the FBD and the kinetic diagram.
- Work–energy: $T_1 + \Sigma U = T_2$, with $T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. Spring energy $\frac{1}{2}kx^2$, gravity mgh, friction work μNd.
- Impulse–momentum: $mv_1 + \int F\,dt = mv_2$. Impact: momentum is conserved; $e = (v_2' - v_1')/(v_1 - v_2)$.
- Rolling without slip: $a = g\sin\theta/(1 + I/(mr^2))$. Hoop: $I = mr^2$. Solid disk: $\frac{1}{2}mr^2$. Sphere: $0.4mr^2$.
- Mechanisms: mechanical advantage, linkages (Grashof), cams, and gear/belt speed ratios.

2. Essential formulas

Name	Formula	Variables / units
Kinematics	$v^2 = v_0^2 + 2as\ ;\ s = v_0t + \frac{1}{2}at^2$	
Newton	$\Sigma F = ma\ ;\ \Sigma M = I\alpha$	
Work–energy	$\frac{1}{2}mv_1^2 + \Sigma U = \frac{1}{2}mv_2^2$	
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$	
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$	
Centripetal	$F = mr\omega^2 = mv^2/r$	
Braking distance	$d = v^2/(2\mu g)$	

3. Handbook search keywords

kinematics · Newton's second law · work-energy · impulse momentum · coefficient of restitution · rolling without slipping · mass moment of inertia

4. Worked examples

Example 1: Crate on an incline
A crate is released on a 25° incline with $\mu_k = 0.15$. Find a and the speed after 4 m.

1. $a = g(\sin 25° - 0.15\cos 25°) = 2.81$ m/s².
2. $v = \sqrt{(2as)} = \sqrt{(2 \cdot 2.81 \cdot 4)} = 4.74$ m/s.

Answer: a = 2.81 m/s², v = 4.74 m/s

Example 2: Impact
A 3 kg ball at 12 m/s hits a stationary 8 kg ball, e = 0.8. Find v_2'.

1. Momentum: $36 = 3v_1' + 8v_2'$. Restitution: $v_2' - v_1' = 9.6$.
2. $v_2' = m_1v_1(1 + e)/(m_1 + m_2) = 3(12)(1.8)/11 = 5.89$ m/s.

Answer: 5.89 m/s

Example 3: Spinning up a rotor
I = 4 kg·m², net torque 50 N·m. Time to reach 1,200 rpm?

1. $\omega = 1{,}200 \cdot 2\pi/60 = 125.7$ rad/s; $\alpha = T/I = 12.5$ rad/s².
2. $t = \omega/\alpha = 10.05$ s.

Answer: 10.1 s

5. Common traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).
- Getting the friction direction wrong (it opposes relative motion).

5.F Basic Machines and Mechanisms (Dynamics): one-page cheat summary

- ► $v^2 = v_0^2 + 2as$
- ► $\Sigma F = ma$, $\Sigma M = I\alpha$
- ► $T = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$
- ► e = sep/approach
- ► Rolling $a = g\sin\theta/(1 + I/mr^2)$
- ► $\omega = 2\pi n/60$

Formulas

Kinematics	$v^2 = v_0^2 + 2as$; $s = v_0t + \frac{1}{2}at^2$
Newton	$\Sigma F = ma$; $\Sigma M = I\alpha$
Work–energy	$\frac{1}{2}mv_1^2 + \Sigma U = \frac{1}{2}mv_2^2$
Restitution	$e = (v_2' - v_1')/(v_1 - v_2)$
Rolling	$a = g\sin\theta/(1 + k^2/r^2)$
Centripetal	$F = mr\omega^2 = mv^2/r$
Braking distance	$d = v^2/(2\mu g)$

Search keywords

kinematics, Newton's second law, work-energy, impulse momentum, coefficient of restitution, rolling without slipping, mass moment of inertia

Top traps

- Forgetting the rotational KE of rolling bodies.
- Treating rpm as rad/s.
- Assuming energy is conserved in an impact (only momentum is).